

Einstein's General Theory of Relativity

Lecture 7: Curvature Tensors, Conservation, and Matter

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Chapter 1

Curvature Tensors, Conservation, and Matter

Overview. A small closed loop turns curvature into an operational measurement: parallel transport returns a vector changed by a linear map. In more than two dimensions that map must remember both the plane of the loop and the vector being transported, which leads to the Riemann tensor and its Ricci contraction. We then turn from geometry to matter. Charge conservation motivates four-currents, energy and momentum lead to the stress–energy tensor, and the geodesic equation closes the chain from matter to curvature to motion.

1.1 Curvature as a Local Transport Experiment

Begin with the two-dimensional picture. Let a vector tangent to a surface be transported parallel to itself around a closed curve. If the enclosed region is flat and the loop is contractible, the vector returns unchanged. If the region is curved, the final vector is generally rotated through an angle $\Delta\theta$.

The loop should be small compared with the scale on which the geometry changes. Curvature is local: an observer need not circumnavigate the Earth to discover that its surface is curved. With sufficiently accurate rulers, protractors, and transport experiments, a tiny neighborhood is enough.

Question from the audience. Must the closed curves be small? ^a

Response. For the local curvature measurement, yes. We shrink the loop until its enclosed area is infinitesimal. Large loops can mix local curvature with global topology, whereas the infinitesimal limit isolates the geometry at the base point.

^aLecture timestamp: 00:02:42–00:03:51.

Nothing in this experiment requires an embedding space. Imagine a bug confined to the surface. Its light rays, rulers, and protractors all lie within the surface, so it can compare lengths, angles, and transported directions without ever looking outward. These are *intrinsic* measurements. How the surface bends in some surrounding space is *extrinsic* information and is not part of the present definition.

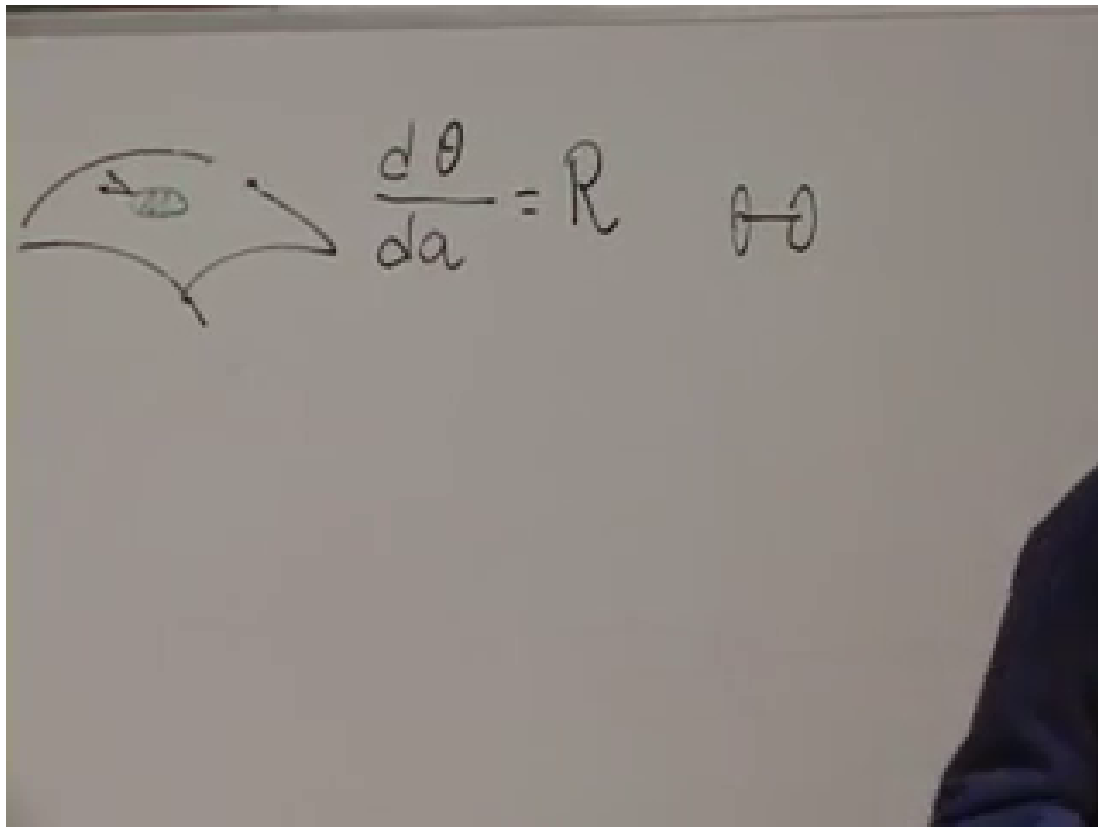


Figure 1.1: The surface loop and the board relation $d\theta/da = R$. In these notes the two-dimensional coefficient is denoted by K , reserving R for curvature tensors and their contractions.

Lecture frame: 00:11:47.

Question from the audience. The drawing shows a surface embedded in three dimensions. Is the curvature being defined by that embedding? ^a

Response. No. The drawing helps our limited visual intuition, but the experiment belongs entirely to the surface. A creature confined there can detect the curvature with tangent rulers and parallel transport. The same intrinsic construction makes sense for a space-time that is not presented as a hypersurface in anything larger.

^aLecture timestamp: 00:03:53–00:06:03.

A rigid two-wheel cart gives a mechanical picture of intrinsic straight motion. Equal wheels constrained not to turn relative to one another roll straight on a plane and follow a geodesic on a curved surface. A gyroscope gives another picture: its axis supplies a direction that can be carried while the base point moves. Neither analogy replaces the connection, but both make the transport experiment concrete.

Choose an orientation for the loop and for positive vector rotation. In two dimensions the leading small-loop law is

$$\Delta\theta = K(p) \Delta A + O((\Delta A)^{3/2}), \quad (1.1)$$

where K is Gaussian curvature and ΔA is oriented area. Reversing the loop reverses $\Delta\theta$. A sphere has positive curvature under the usual convention; a saddle has negative curvature.

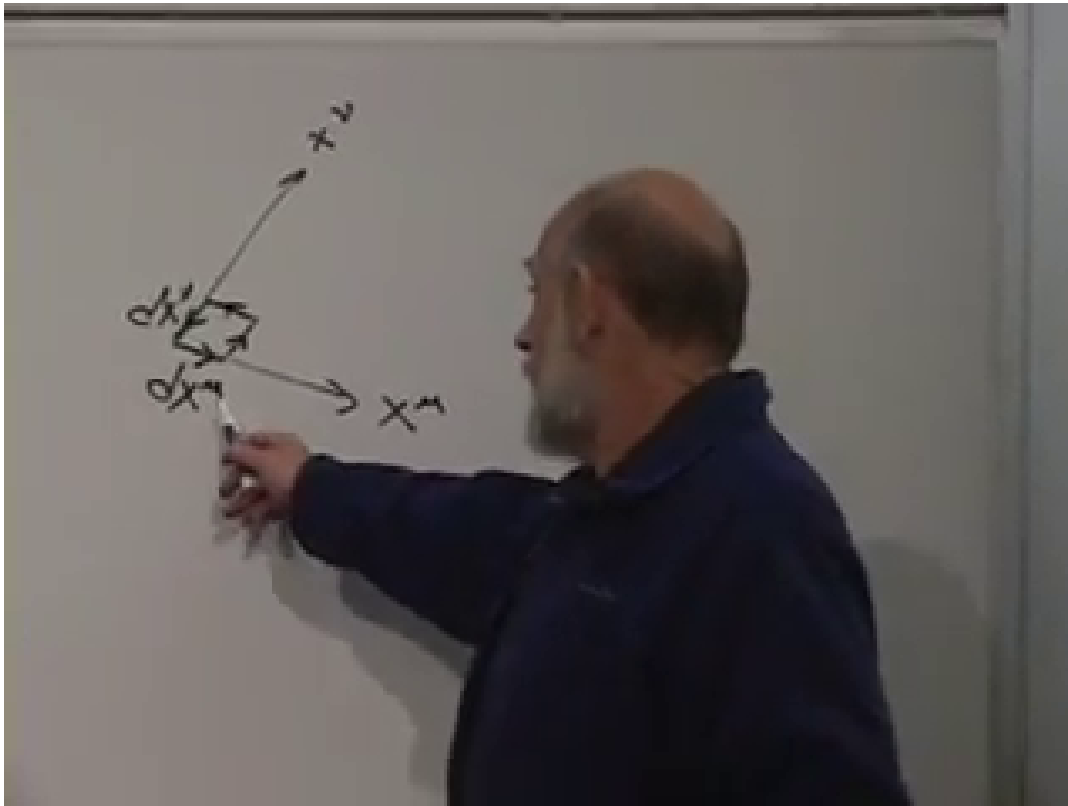


Figure 1.2: An infinitesimal coordinate rectangle in the x^μ - x^ν plane. Its two directed edges identify the plane and orientation on which the curvature map is evaluated.

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1.2 Two Planes in More Than Two Dimensions

On a surface there is only one tangent plane, so the rotation angle nearly contains the whole story. In $D > 2$, two distinct geometric choices appear.

First, the infinitesimal loop lies in a plane selected by two displacement directions, say x^μ and x^ν . Second, parallel transport acts on the initial vector and may rotate or mix its components in another plane. In three dimensions one may replace each oriented plane by a dual axis, but that shortcut is special to three dimensions. In arbitrary dimension the pair of directions must be kept explicitly.

Transporting an orthonormal gyroscope triad illustrates the second plane. The triad may return rotated about an axis unrelated to the axis normal to the loop. The curvature therefore cannot be represented by one scalar in general.

Question from the audience. Can parallel transport destroy perpendicularity between two vectors? ^a

Response. Not for the Levi-Civita connection. Metric compatibility gives

$$\frac{D}{Ds}(g_{\rho\sigma}U^\rho V^\sigma) = 0$$

when both vectors are parallel transported. Lengths and mutual angles, including a right angle, are preserved. More general affine connections need not have this property.

^aLecture timestamp: 00:16:15–00:17:30.

1.3 The Infinitesimal Rectangle

Let V^ρ be transported along a curve. Parallel transport is defined by

$$\frac{DV^\rho}{Ds} = \frac{dV^\rho}{ds} + \Gamma^\rho_{\sigma\mu} V^\sigma \frac{dx^\mu}{ds} = 0. \quad (1.2)$$

For one infinitesimal displacement this becomes

$$dV^\rho = -\Gamma^\rho_{\sigma\mu} V^\sigma dx^\mu. \quad (1.3)$$

Now follow the four sides of a genuinely closed infinitesimal rectangle with edge vectors ϵ^μ and η^ν . Contributions that would occur on a path followed immediately by its reverse cancel. Neighboring sides do not cancel completely because the connection on the second side is evaluated at a displaced point, and because successive connection matrices need not commute.

The surviving change is bilinear in the two edge vectors and linear in the initial vector:

$$\delta V^\rho = R^\rho_{\sigma\mu\nu} V^\sigma \epsilon^\mu \eta^\nu + O(\ell^3). \quad (1.4)$$

Here ℓ is a characteristic edge length. The antisymmetric combination

$$\Sigma^{\mu\nu} = \epsilon^\mu \eta^\nu - \epsilon^\nu \eta^\mu \quad (1.5)$$

is the oriented area bivector. It is more precise than calling the area the simple product of two edge lengths, since the coordinate directions need not be orthogonal or normalized.

Question from the audience. One index was used twice for different roles. Which indices belong to the vector, and which belong to the loop? ^a

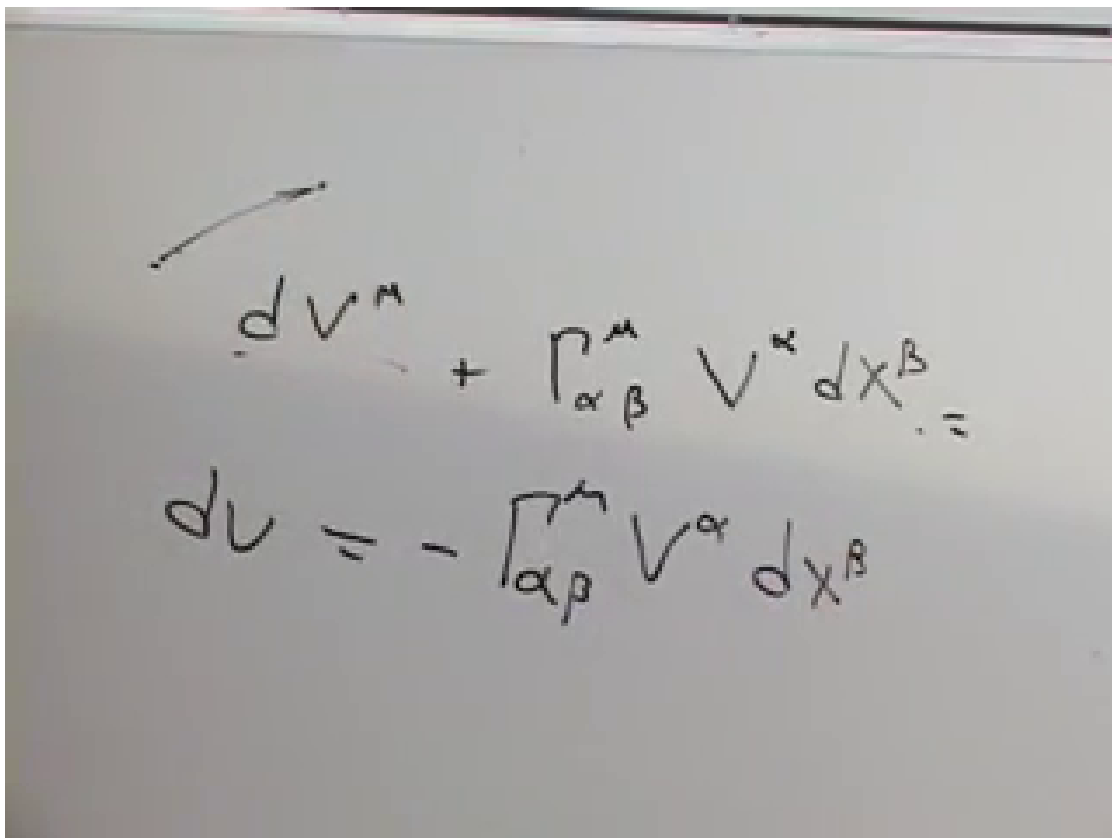
Response. Use four distinct slots. In $R^\rho_{\sigma\mu\nu}$, the index σ selects the initial component of V , ρ selects the resulting component, and μ, ν identify the oriented loop plane. Repeated indices are summed only after those roles have been separated.

^aLecture timestamp: 00:20:57–00:21:10.

Question from the audience. Are the two infinitesimal displacements the same quantity, and are their indices summed? ^a

Response. They are the two independently chosen edge vectors of the small loop. Their components contract with the last two curvature indices. Writing $\epsilon^\mu \eta^\nu$, or equivalently the area bivector $\Sigma^{\mu\nu}$, prevents the two directions from being mistaken for one displacement.

^aLecture timestamp: 00:36:31–00:37:40.



The image shows a chalkboard with two equations written in black ink. The first equation is $dV^\mu + \Gamma_{\alpha\beta}^\mu V^\alpha dx^\beta = 0$, with a small arrow pointing to the left above the dV^μ term. The second equation is $dV = -\Gamma_{\alpha\beta}^\mu V^\alpha dx^\beta$.

Figure 1.3: The board form of the infinitesimal parallel-transport law. The connection changes vector components as the base point moves.

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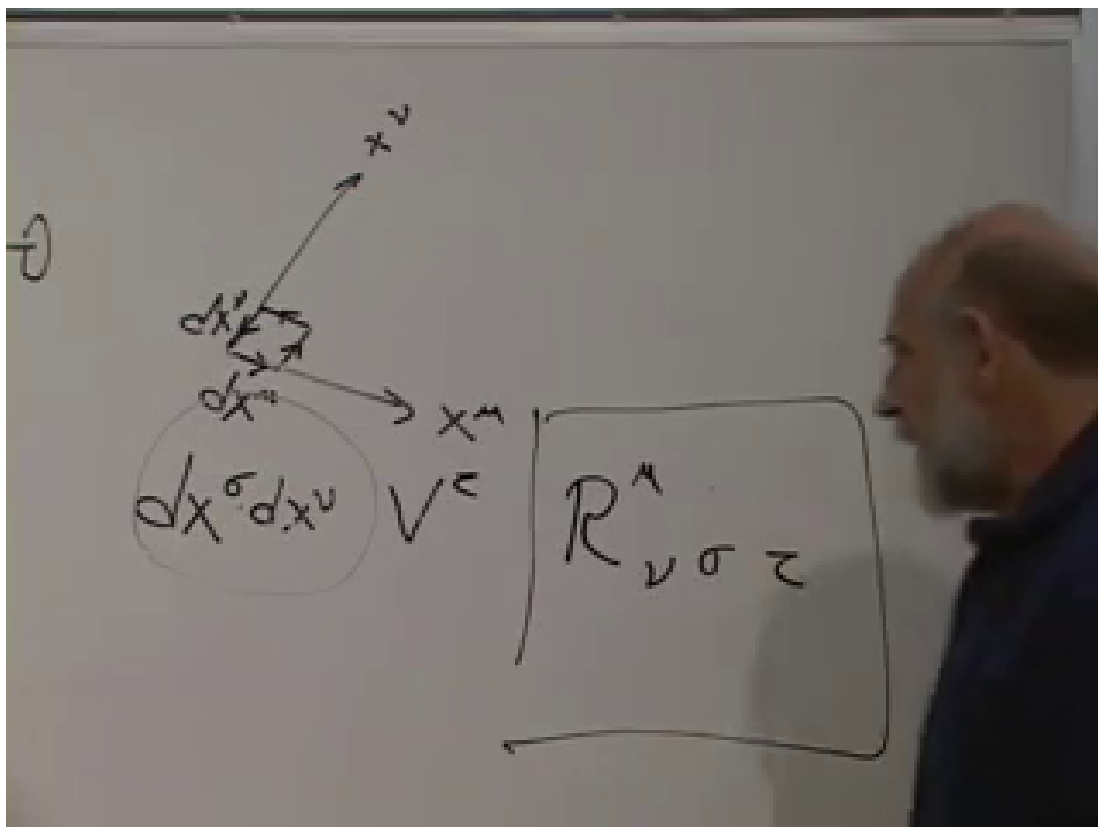


Figure 1.4: The board separates the two displacement directions, the transported vector component, and the four-index curvature coefficient. Together they form the linear holonomy map in Eq. (1.4).

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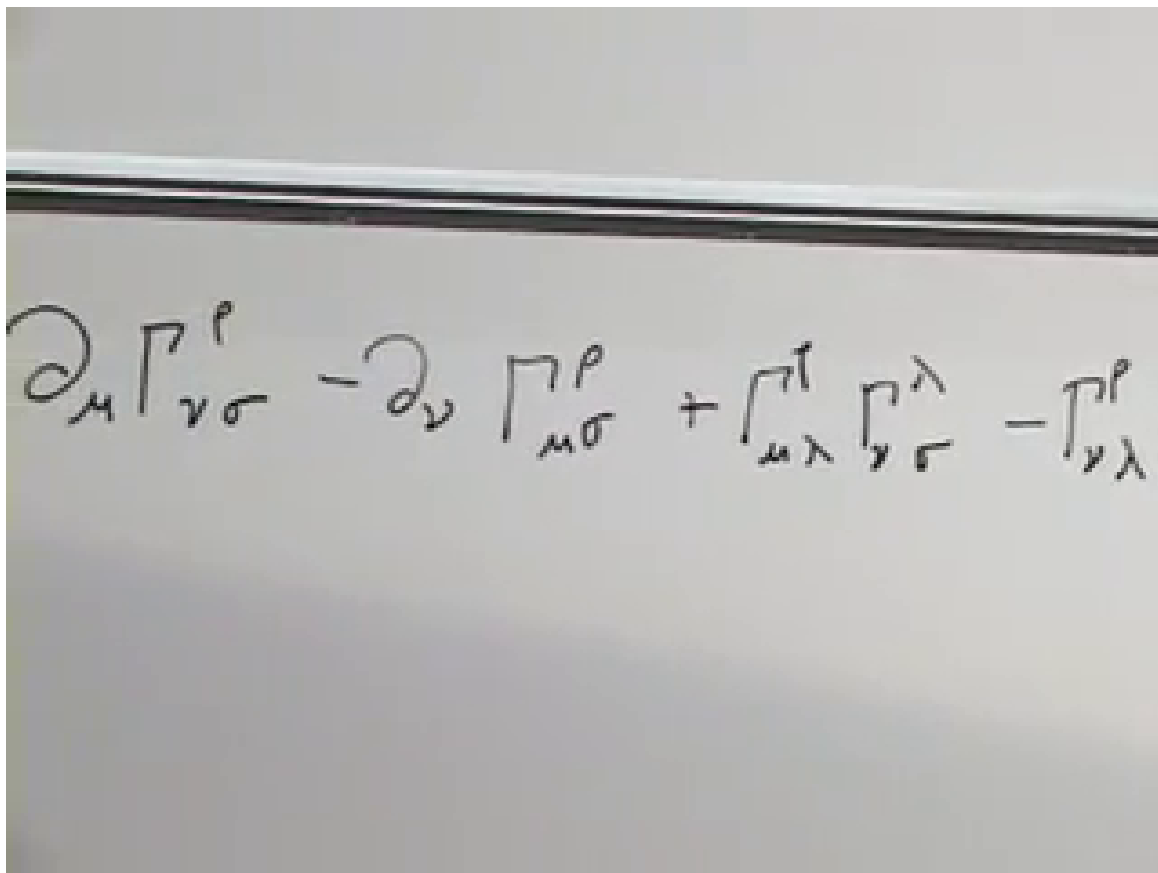


Figure 1.5: The camera follows the completed Riemann expression across the board; this frame records its derivative and ordered quadratic terms. Equation (1.7) reconstructs the full line.

Lecture frame: 00:32:23.

1.4 The Riemann Curvature Tensor

We fix the convention

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho_{\sigma\mu\nu}V^\sigma. \quad (1.6)$$

With this convention,

$$\begin{aligned} R^\rho_{\sigma\mu\nu} &= \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} \\ &\quad + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}. \end{aligned} \quad (1.7)$$

Changing the order in the commutator or reversing the loop orientation changes the overall sign. Once a convention is declared, every later formula must follow it consistently.

Although individual Christoffel symbols are not tensor components, the combination in Eq. (1.7) is tensorial. The geometric reason is more illuminating than a long transformation-law calculation. Transport around the loop starts and ends at the same point, so δV^ρ is the difference of two vectors in one tangent space. Equation (1.4) holds for every initial vector and every oriented infinitesimal area. Its multilinear coefficient must therefore transform as a tensor.

$$+ \sqrt{\Gamma_{\mu\lambda} \Gamma_{\nu\sigma}} - \Gamma_{\nu\lambda} \Gamma_{\mu\sigma}$$

$$\sqrt{\Gamma_{\mu\nu} \Gamma_{\lambda\sigma}} - \Gamma_{\nu\mu} \Gamma_{\lambda\sigma}$$

Figure 1.6: The two quadratic connection products are highlighted in opposite order. Their difference is the commutator term in the curvature.

Lecture frame: 00:36:08.

Question from the audience. Why is the curvature coefficient a tensor when it is built from non-tensorial Christoffel symbols? ^a

Response. The non-tensorial pieces cancel in precisely the combination above. Operationally, the closed-loop change is a vector at the original point, and it depends linearly on the initial vector and the two edge vectors. That coordinate-independent multilinear map is the Riemann tensor.

^aLecture timestamp: 00:32:42–00:34:44.

For each fixed direction μ , regard $(\Gamma_\mu)^\rho{}_\sigma = \Gamma^\rho{}_{\mu\sigma}$ as a matrix. Then the quadratic terms are a matrix commutator:

$$R_{\mu\nu} = \partial_\mu \Gamma_\nu - \partial_\nu \Gamma_\mu + [\Gamma_\mu, \Gamma_\nu]. \quad (1.8)$$

Because matrix multiplication is not generally commutative, the two orders give distinct terms.

Question from the audience. Are the quadratic terms literally a commutator? ^a

Response. Yes. Once the first and last tensor slots are treated as matrix indices, the products are $\Gamma_\mu \Gamma_\nu - \Gamma_\nu \Gamma_\mu$. Together with the derivative terms they form the commutator of covariant derivatives in Eq. (1.6).

^aLecture timestamp: 00:34:44–00:36:31.

1.4.1 Normal coordinates do not remove curvature

At any regular point p , normal coordinates can be chosen so that

$$g_{\mu\nu}(p) = \eta_{\mu\nu}, \quad \partial_\lambda g_{\mu\nu}(p) = 0, \quad \Gamma^\rho_{\mu\nu}(p) = 0. \quad (1.9)$$

The second derivatives of the metric do *not* generally vanish. They contain the first locally invariant information about curvature. At p , Eq. (1.7) loses its quadratic connection terms, but derivatives of the connection remain. This is why gravity can be removed at one event while tidal effects cannot.

Question from the audience. How would an observer actually keep a vector parallel while moving over a curved Earth? ^a

Response. One may carry a gyroscope, or repeatedly compare the vector with a locally inertial orthonormal frame. In coordinates the same instruction is $DV^\rho/Ds = 0$. A tangent transported along a geodesic continues straight; other vectors are transported by the same connection. Normal coordinates simplify the rule at one point, not along the whole route.

^aLecture timestamp: 00:37:49–00:42:49.

1.5 Symmetries and Independent Components

Lower the first index with the metric:

$$R_{\alpha\beta\mu\nu} = g_{\alpha\rho} R^\rho_{\beta\mu\nu}. \quad (1.10)$$

For the torsion-free Levi-Civita connection,

$$R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu}, \quad (1.11)$$

$$R_{\alpha\beta\mu\nu} = -R_{\alpha\beta\nu\mu}, \quad (1.12)$$

$$R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}, \quad (1.13)$$

$$R_{\alpha[\beta\mu\nu]} = 0. \quad (1.14)$$

The last line is the first Bianchi identity. The tensor is antisymmetric within each index pair, symmetric under exchange of the pairs, and subject to one cyclic identity. It is not assigned a sign by an arbitrary permutation of all four indices.

Question from the audience. Does reversing the loop reverse the transport effect? ^a

Response. Yes. Reversing the oriented area exchanges μ and ν , and $R^\rho_{\sigma\mu\nu} = -R^\rho_{\sigma\nu\mu}$. The leading holonomy therefore changes sign.

^aLecture timestamp: 00:44:52–00:45:20.


$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma^{\rho}{}_{\nu\sigma} -$$
$$R_{\lambda\sigma\mu\nu} = g_{\lambda\rho} R^{\rho}{}_{\sigma\mu\nu}$$

Figure 1.7: Lowering the output index produces the fully covariant Riemann tensor, the form in which its pair symmetries are easiest to state.

Lecture frame: 00:44:08.

$$+ \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda$$

$$- R_{\beta\alpha\gamma\delta} = -R_{\alpha\beta\gamma\delta}$$

Figure 1.8: The board develops antisymmetry within an index pair. The completed relations are Eqs. (1.11)–(1.14).

Lecture frame: 00:47:48.

Question from the audience. Can every permutation be handled by attaching the sign of the permutation, and how many independent components remain? ^a

Response. No. Only the stated pair operations have simple signs, and the Bianchi identity supplies an additional relation. In D dimensions the number of independent components is

$$N_R = \frac{D^2(D^2 - 1)}{12}.$$

Thus $N_R = 1$ in two dimensions, 6 in three dimensions, and 20 in four dimensions. The tentative board count is corrected by including all pair symmetries and the cyclic identity.

^aLecture timestamp: 00:48:38–00:52:52.

For a two-sphere of radius a ,

$$K = \frac{1}{a^2}. \quad (1.15)$$

The unique independent curvature component in two dimensions is therefore fixed by one scalar function.

Question from the audience. How does the sphere's radius enter the curvature? ^a

Response. The Gaussian curvature is $1/a^2$. A larger sphere looks flatter on a fixed small patch, and the same small area produces a smaller holonomy angle.

^aLecture timestamp: 00:54:12–00:54:57.

The infinitesimal path must actually close. A coordinate parallelogram is only a convenient leading-order construction. For the torsion-free connection used here, one may choose a genuinely closed small loop whose shape corrections change only higher-order terms; the leading area-curvature relation remains Eq. (1.4).

Question from the audience. What if the four prescribed displacements fail to close exactly? ^a

Response. Define a closed parameterized loop first, then expand its holonomy. In a torsion-free geometry the small coordinate rectangle can be corrected at higher order without changing the leading $R \cdot \Sigma$ term. An open path would compare vectors at different points and would not by itself define the same local curvature measurement.

^aLecture timestamp: 00:55:01–00:57:40.

1.6 The Ricci Contraction

One contraction of the Riemann tensor is especially important:

$$R_{\beta\nu} = R^\alpha_{\beta\alpha\nu} = g^{\alpha\mu} R_{\alpha\beta\mu\nu}. \quad (1.16)$$

For the Levi-Civita connection, the Ricci tensor is symmetric:

$$R_{\beta\nu} = R_{\nu\beta}. \quad (1.17)$$

Contracting once more gives the Ricci scalar,

$$R = g^{\beta\nu} R_{\beta\nu}. \quad (1.18)$$

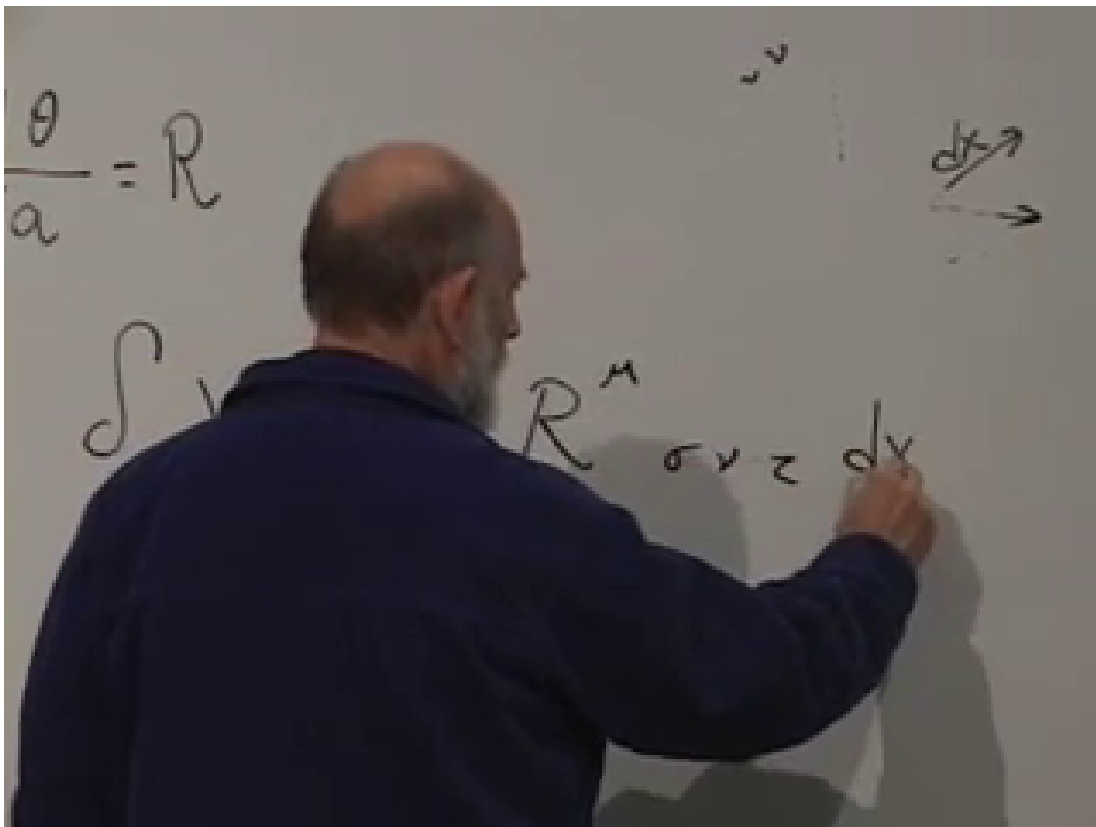


Figure 1.9: The board returns to the small-loop relation: the curvature tensor contracts with two directed edge elements and the transported vector to give the leading vector change.

Lecture frame: 00:53:23.

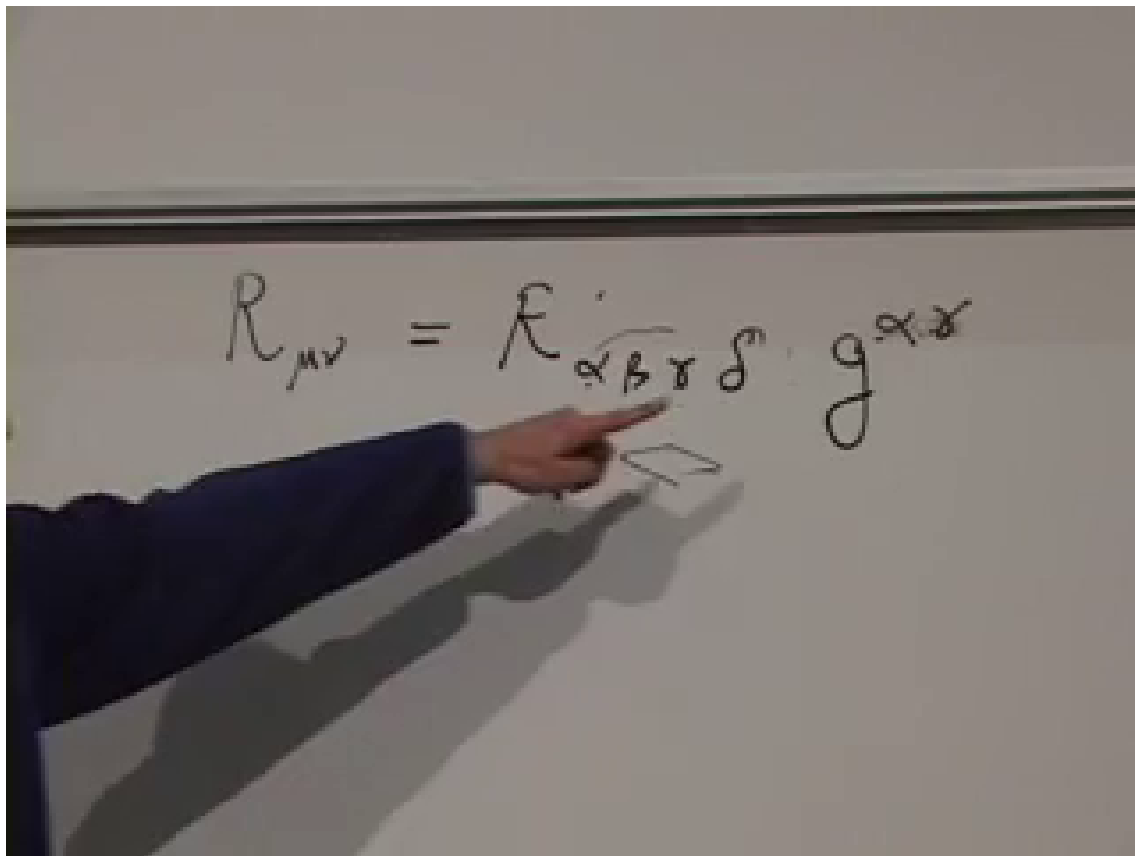
A photograph of a whiteboard with a hand pointing to a mathematical equation. The equation is $R_{\mu\nu} = R^{\alpha}{}_{\beta\gamma\delta} g^{\gamma\delta}$. The hand is pointing to the index γ in the Riemann tensor, which is part of a contraction with the metric tensor $g^{\gamma\delta}$. The Riemann tensor is written as $R^{\alpha}{}_{\beta\gamma\delta}$ with a dot over the α index. The metric tensor is written as $g^{\gamma\delta}$. The hand is pointing to the γ index in the Riemann tensor, which is the index being contracted with the γ index in the metric tensor. The equation is written in black ink on a white background. The hand is wearing a dark blue sleeve. The background is a plain white wall.

Figure 1.10: The Ricci tensor is obtained by contracting one index from each antisymmetric pair of the Riemann tensor.

Lecture frame: 01:02:23.

In three dimensions the Riemann tensor contains no information beyond the Ricci tensor. Explicitly,

$$R_{abcd} = g_{ac}R_{bd} - g_{ad}R_{bc} - g_{bc}R_{ad} + g_{bd}R_{ac} - \frac{R}{2}(g_{ac}g_{bd} - g_{ad}g_{bc}). \quad (1.19)$$

In four dimensions this is no longer true; the Weyl tensor carries the remaining curvature information.

Question from the audience. What geometric problem does the Ricci tensor solve? ^a

Response. It is the natural trace of the full curvature map. In three dimensions it determines all of the Riemann tensor, as Eq. (1.19) shows. In four-dimensional space-time it does not determine every tidal degree of freedom, but it is the contraction that enters the Einstein tensor and hence the field equation.

^aLecture timestamp: 01:03:31–01:05:27.

1.7 From Geometry to Matter

We have reached the geometric half of general relativity. Curvature is the coordinate-independent obstruction measured by small-loop transport and, physically, by tidal relative acceleration. The other half asks what produces that curvature.

The answer cannot be mass density alone. Special relativity mixes energy and momentum between frames, so a relativistic source must contain local densities and fluxes of both. Before constructing it, we use electric charge as a simpler conserved quantity.

1.8 Charge Density and Current

For a small spatial volume ΔV containing charge ΔQ , define

$$\rho_q = \lim_{\Delta V \rightarrow 0} \frac{\Delta Q}{\Delta V}. \quad (1.20)$$

The charge in a region Ω is

$$Q_\Omega(t) = \int_\Omega \rho_q(t, \mathbf{x}) d^3x. \quad (1.21)$$

Imagine the fixed spatial region extended through time. It becomes a world-tube. Charged particles trace worldlines through that tube. Charge inside the spatial box may change because worldlines cross its side boundary, even though the total charge of the complete isolated system is constant. Under local conservation, charge worldlines do not simply end at one event and begin at an unrelated distant event.

The current component j^i measures charge crossing an oriented area normal to the i -direction per unit area per unit time:

$$j^i = \frac{dQ}{dA_i dt}. \quad (1.22)$$

When $c = 1$, time and distance have the same units, so charge density and current density fit naturally into one four-vector,

$$J^\mu = (\rho_q, j^1, j^2, j^3). \quad (1.23)$$

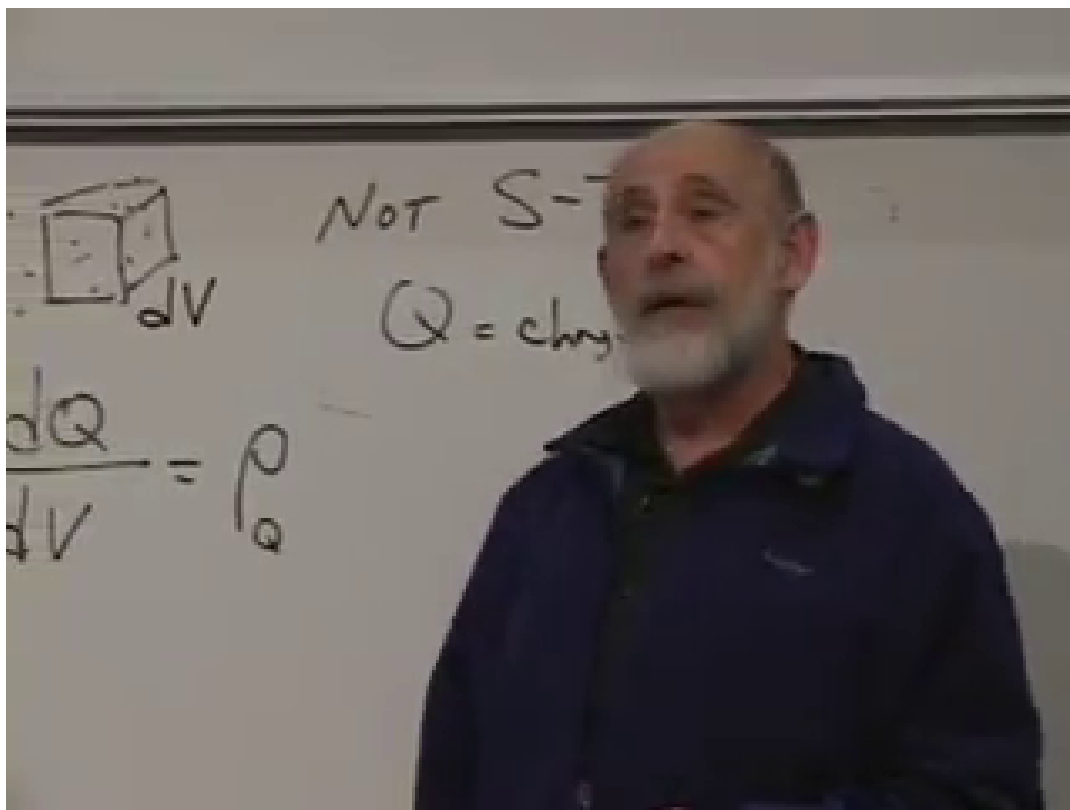


Figure 1.11: A small volume element and the local definition $\rho_q = dQ/dV$. Charge itself is not a space-time component; its density depends on the spatial slice.

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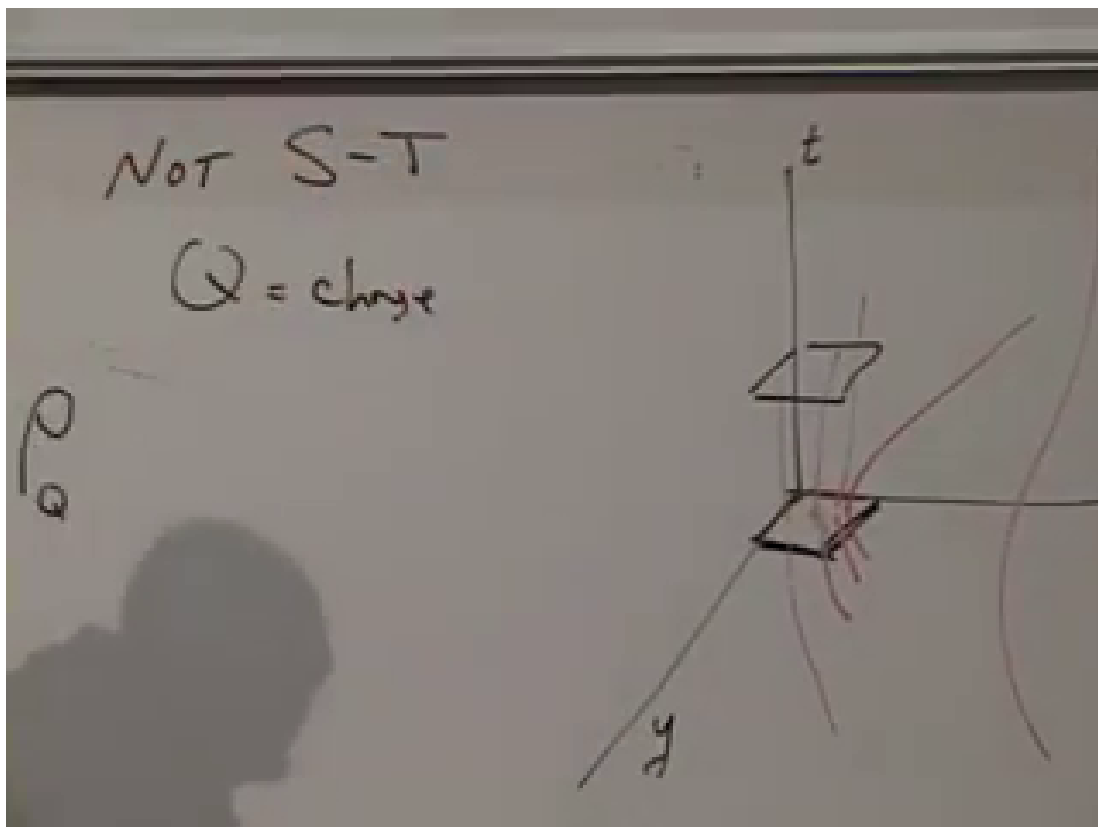


Figure 1.12: A spatial box extended in time. Charge worldlines cross the boundary, turning change of enclosed charge into a flux statement.

Lecture frame: 01:11:53.

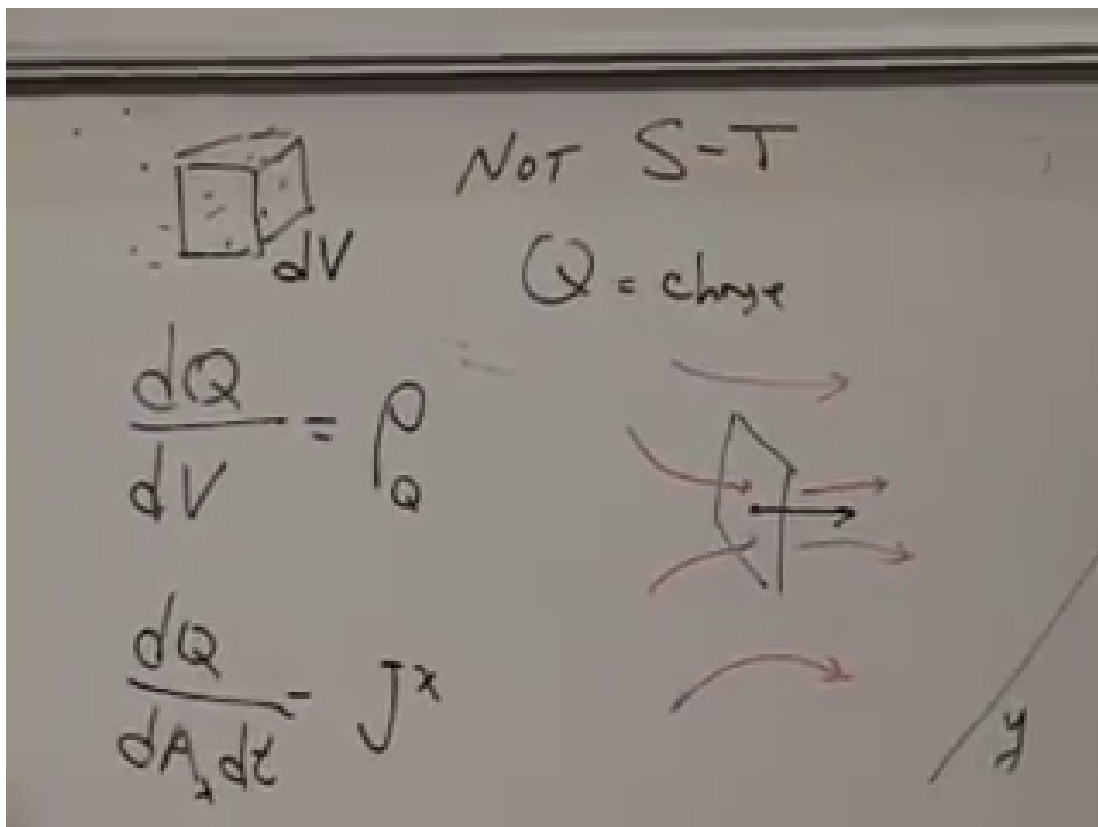


Figure 1.13: The board definitions of charge density and current through an oriented window, accompanied by the flux picture.

Lecture frame: 01:17:08.

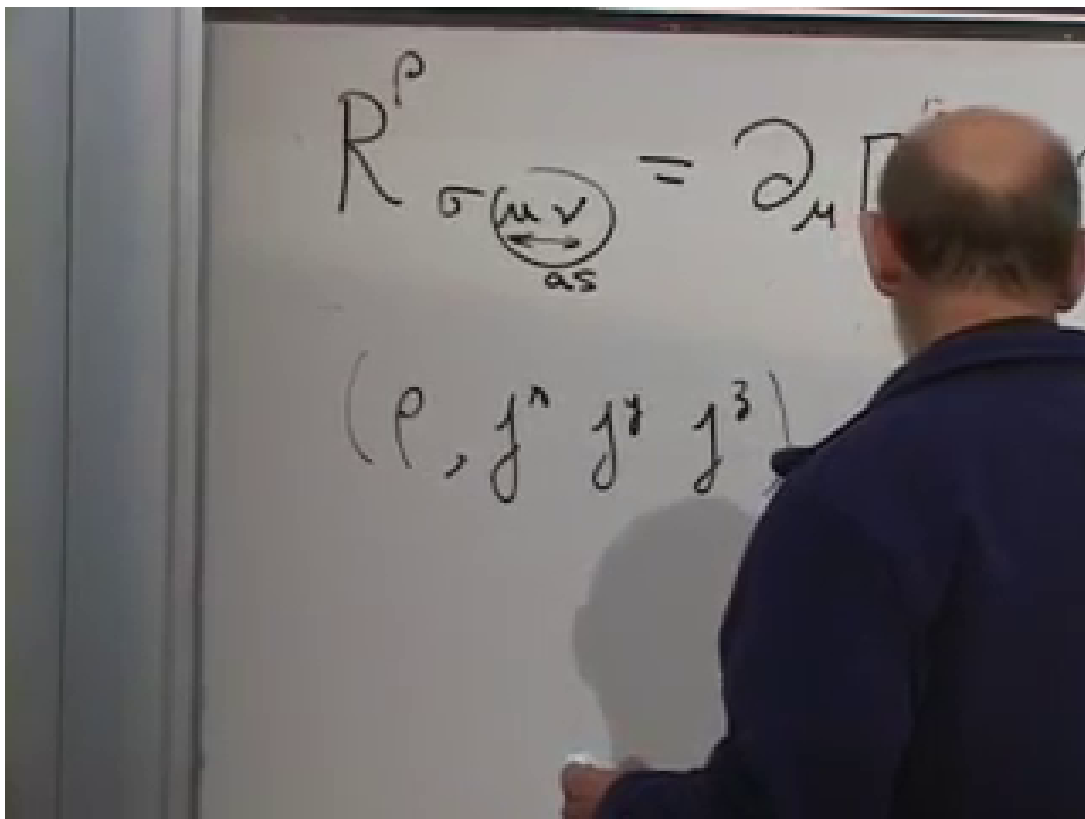


Figure 1.14: The four-current components $J^{\mu} = (\rho_q, j^1, j^2, j^3)$. A Lorentz transformation mixes density and spatial current.

Lecture frame: 01:17:53.

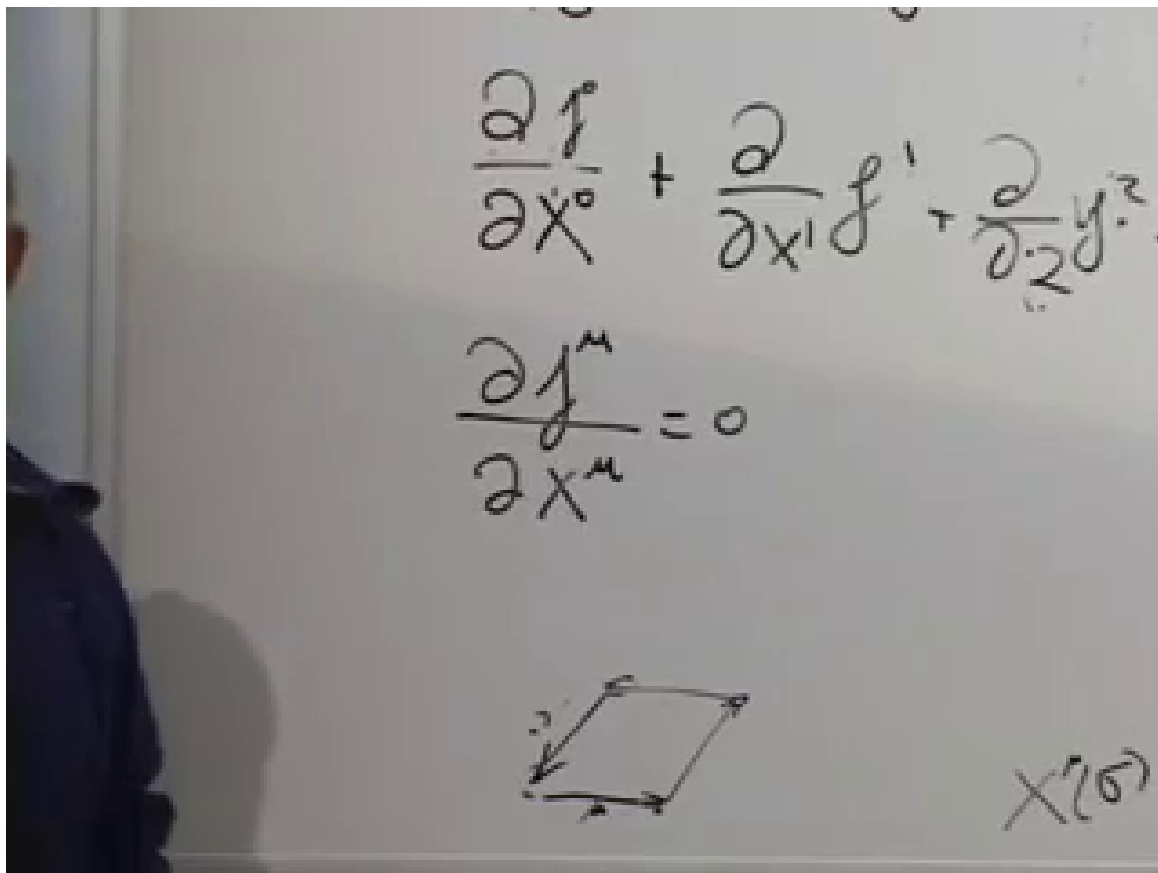


Figure 1.15: The continuity equation in three-dimensional and four-vector form. A uniform current through a box has equal inflow and outflow, so the enclosed density remains unchanged.

Lecture frame: 01:24:38.

1.9 Local Conservation

Global conservation says that the total charge of an isolated system does not change. Local conservation says more: charge in a region changes only because current flows through its boundary. Without locality, charge could disappear here and reappear far away while preserving only the global total.

For a fixed volume Ω ,

$$\frac{d}{dt} \int_{\Omega} \rho_q d^3x = - \oint_{\partial\Omega} \mathbf{j} \cdot d\mathbf{A}. \quad (1.24)$$

The divergence theorem gives the differential continuity equation,

$$\boxed{\frac{\partial \rho_q}{\partial t} + \nabla \cdot \mathbf{j} = 0}. \quad (1.25)$$

In space-time notation,

$$\partial_\mu J^\mu = 0. \quad (1.26)$$

1.9.1 Charge is a scalar; density is not

The total electric charge Q is Lorentz invariant. The density ρ_q is not a scalar: changing frames changes both the spatial volume used to define the density and the motion of the charges. What one observer calls mostly density another may describe as a mixture of density and current. The four-vector J^μ , not ρ_q alone, is the covariant object.

Question from the audience. Is one coulomb of charge necessarily an enormous explosive amount, and what does a one-farad capacitor contain? ^a

Response. A capacitor stores separated charges $+Q$ and $-Q$, so its net charge is zero. The exact relations are

$$Q = CV, \quad U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}.$$

A one-farad capacitor at one volt carries one coulomb in magnitude on each plate and stores 0.5 joule. That is not an explosion. By contrast, concentrating an uncompensated net charge of one coulomb in a small isolated object would require extreme electrostatic conditions. The geometry and voltage matter; the unit of charge alone does not fix the stored energy.

^aLecture timestamp: 01:26:12–01:31:02.

1.10 Energy and Momentum

Energy and momentum are also conserved, but neither is separately frame-independent. They form the four-momentum

$$P^\mu = (E, p^1, p^2, p^3) \quad (c = 1). \quad (1.27)$$

A body at rest in one frame has energy and no spatial momentum; a moving observer assigns it both kinetic energy and momentum. Consequently, local energy conservation and the three local momentum-conservation laws must be components of one covariant statement.

1.11 The Stress–Energy Tensor

The local relativistic source is the stress–energy tensor $T^{\nu\mu}$. We use the lecture’s bookkeeping convention: the first index ν labels the conserved quantity and the second index μ labels the direction of flux. Then

$$T^{00} = \text{energy density}, \quad (1.28)$$

$$T^{0i} = \text{flux of energy in direction } i, \quad (1.29)$$

$$T^{i0} = \text{density of } i\text{-momentum}, \quad (1.30)$$

$$T^{ij} = \begin{array}{l} \text{flux of } i\text{-momentum across a surface} \\ \text{whose normal points in direction } j. \end{array} \quad (1.31)$$

The spatial block T^{ij} is the stress tensor. Its diagonal entries are pressures or normal stresses; its off-diagonal entries are shear stresses. A stream of particles moving obliquely, for example, can carry x -momentum across a surface whose normal points in the y -direction.

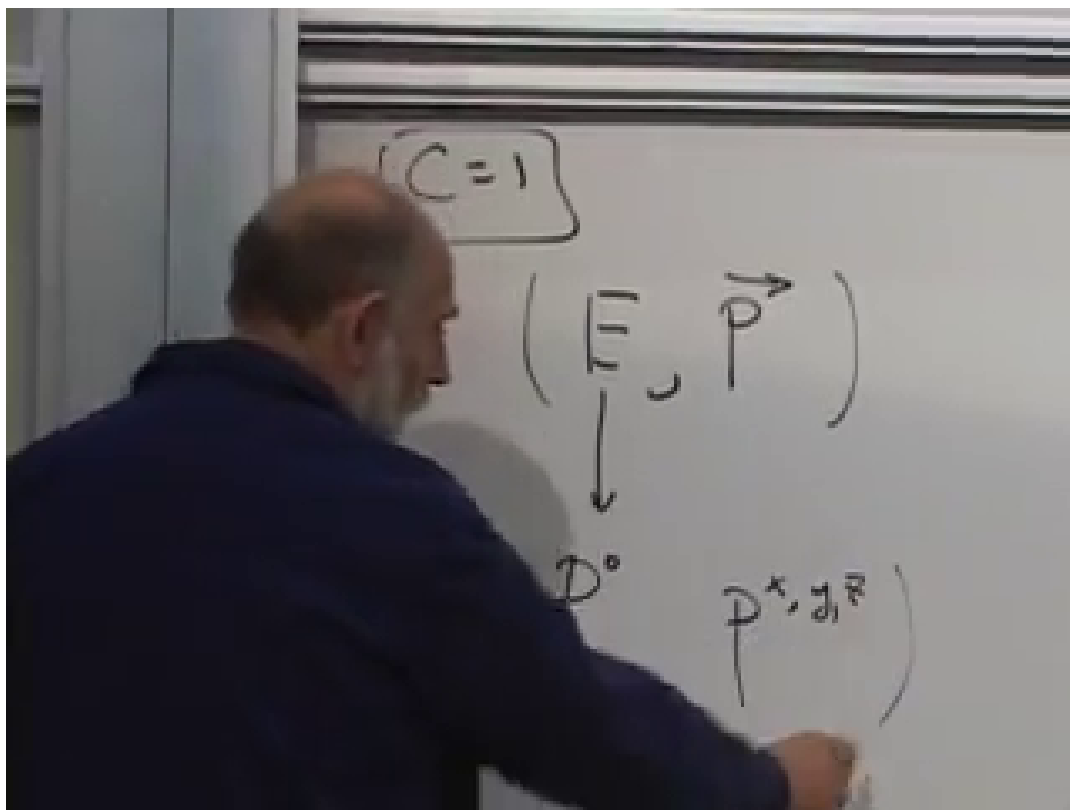


Figure 1.16: Energy and three-momentum assembled into the four-momentum P^μ . Lorentz transformations mix its components.

Lecture frame: 01:33:23.

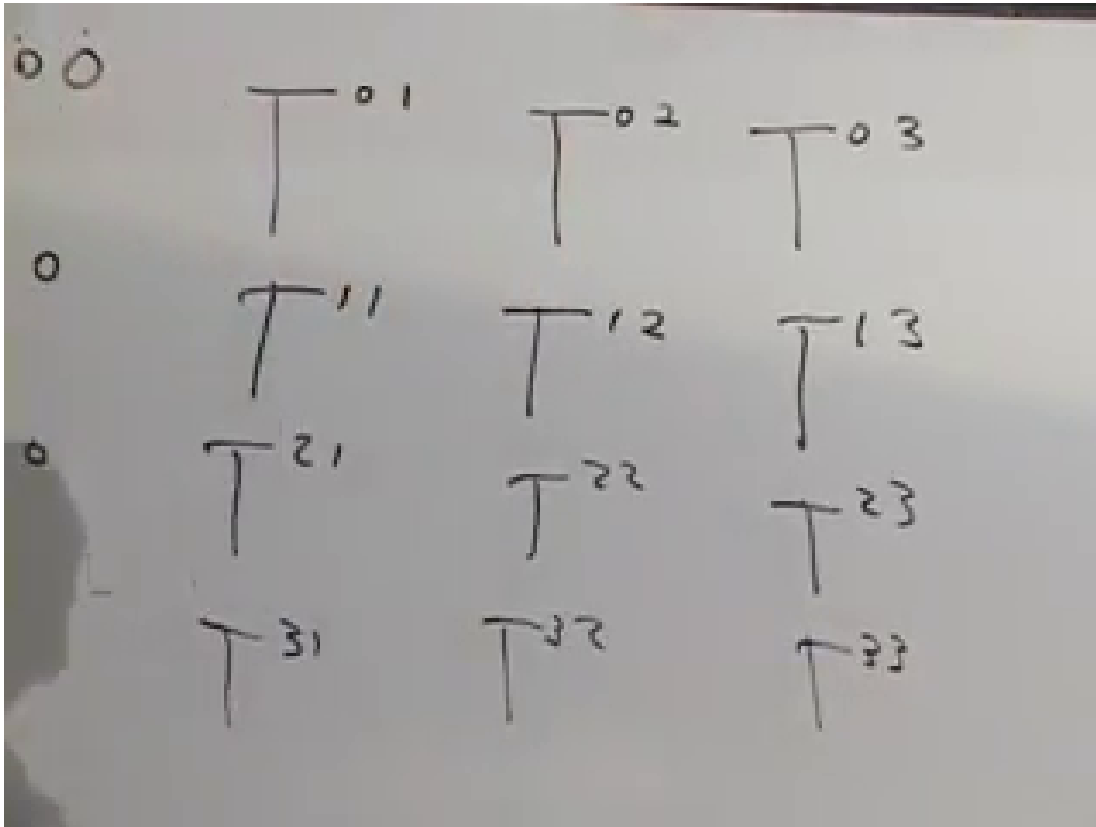


Figure 1.17: The complete 4×4 array of stress–energy components. Rows label energy or a momentum component; columns distinguish density from the three flux directions.

Lecture frame: 01:41:23.

$$T^{\mu\nu} = \begin{pmatrix} T^{00} & T^{01} & T^{02} & T^{03} \\ T^{10} & T^{11} & T^{12} & T^{13} \\ T^{20} & T^{21} & T^{22} & T^{23} \\ T^{30} & T^{31} & T^{32} & T^{33} \end{pmatrix}. \quad (1.32)$$

For the standard symmetric stress–energy tensor used as the gravitational source,

$$T^{\mu\nu} = T^{\nu\mu}. \quad (1.33)$$

Microscopically, energy flux equals momentum density in units $c = 1$. For fields with intrinsic spin, the canonical tensor can first be nonsymmetric; the symmetric, conserved tensor coupled to gravity is obtained by the standard improvement.

Question from the audience. Is the array automatically a tensor merely because its entries describe physical quantities? ^a

Response. Physical meaning motivates the construction but does not by itself prove a transformation law. One proves tensoriality by a covariant microscopic or field-theoretic definition, or by showing how the local density and fluxes transform under Lorentz changes of frame. Once that is done, the component interpretations above follow in every local inertial frame.

^aLecture timestamp: 01:42:23–01:43:00.

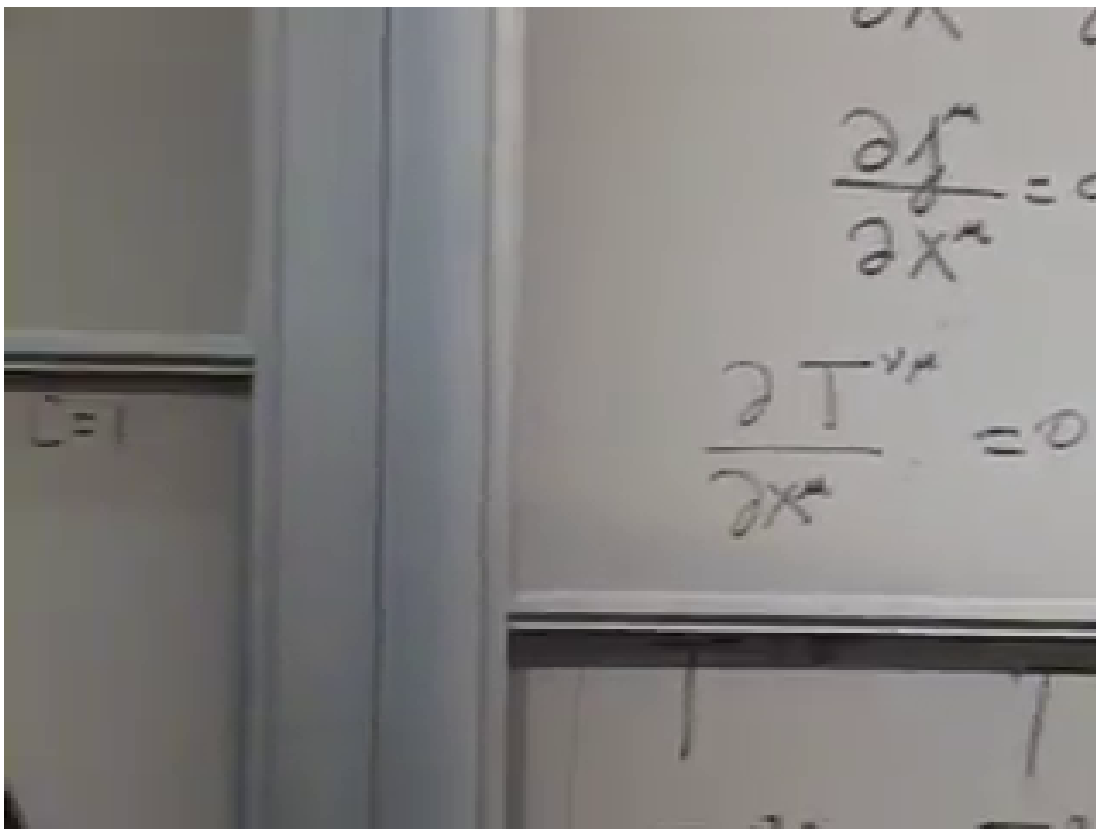


Figure 1.18: The four compact conservation equations for energy and momentum, written as the divergence of $T^{\nu\mu}$.

Lecture frame: 01:46:08.

1.12 Four Local Conservation Laws

Each row obeys a continuity equation. With the index convention just chosen,

$$\frac{\partial T^{\nu 0}}{\partial t} + \frac{\partial T^{\nu 1}}{\partial x} + \frac{\partial T^{\nu 2}}{\partial y} + \frac{\partial T^{\nu 3}}{\partial z} = 0. \quad (1.34)$$

Equivalently,

$$\boxed{\partial_\mu T^{\nu\mu} = 0}. \quad (1.35)$$

If the symmetric convention $T^{\mu\nu} = T^{\nu\mu}$ is already in use, this is the familiar $\partial_\mu T^{\mu\nu} = 0$.

Equation (1.35) is written with ordinary derivatives because we have described a local inertial frame in special relativity. In curved space-time the covariant statement is

$$\nabla_\mu T^{\nu\mu} = 0. \quad (1.36)$$

Question from the audience. Why are ordinary derivatives being used if the goal is general relativity? ^a

Response. The matter tensor is first organized in a local inertial frame, where the connection vanishes at the chosen event. The curved-space equation is obtained by replacing the ordinary divergence with the covariant divergence.

^aLecture timestamp: 01:46:31–01:46:54.

1.13 Curvature, Sources, and Geodesic Motion

The structure of general relativity is now visible even before the field equation is written:

$$T_{\mu\nu} \longrightarrow \text{space-time curvature} \longrightarrow \text{geodesic motion.} \quad (1.37)$$

Mass contributes because $E = mc^2$, but energy density is only one component of the source. Momentum density, energy flux, pressure, and shear must also contribute, because Lorentz transformations mix them.

A freely falling particle follows a space-time geodesic. With proper time τ as an affine parameter,

$$\boxed{\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\sigma} \frac{dx^\nu}{d\tau} \frac{dx^\sigma}{d\tau} = 0}. \quad (1.38)$$

Take the slow-motion, weak-field limit. The spatial components of $dx^\mu/d\tau$ are small compared with the time component, and

$$\frac{dt}{d\tau} \simeq 1. \quad (1.39)$$

For a spatial index i , the dominant term in the geodesic equation is

$$\frac{d^2 x^i}{dt^2} \simeq -\Gamma^i_{00}. \quad (1.40)$$

Question from the audience. Which component of the four-velocity remains large in the nonrelativistic limit? ^a

Response. The time component. With $c = 1$, $dx^0/d\tau = dt/d\tau \simeq 1$, while the three spatial components are the small ordinary velocities. That is why the 00 connection coefficient dominates the spatial geodesic equation.

^aLecture timestamp: 01:53:42–01:53:51.

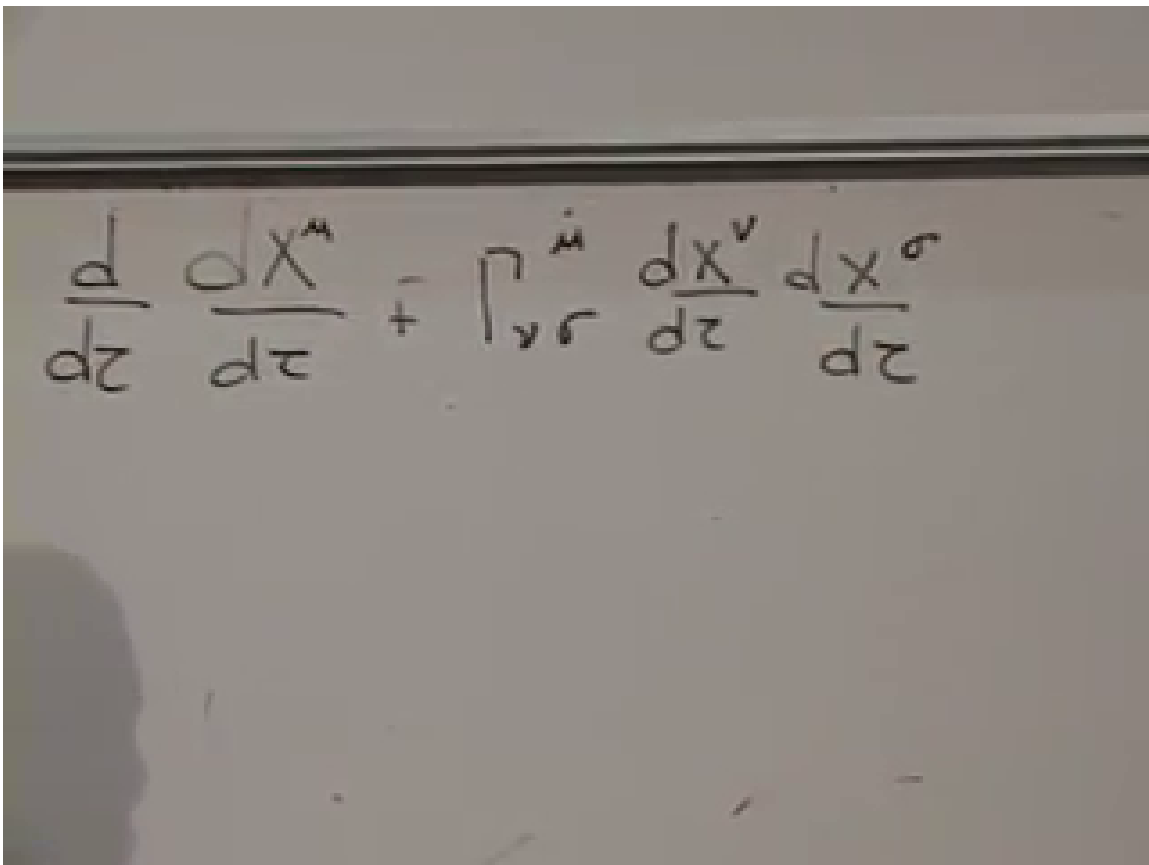
For the weak static metric with signature $(+---)$,

$$g_{00} \simeq 1 + 2\Phi, \quad \Gamma^i_{00} \simeq \partial_i \Phi, \quad (1.41)$$

so Eq. (1.40) becomes

$$\frac{d^2 \mathbf{x}}{dt^2} \simeq -\nabla \Phi. \quad (1.42)$$

The next step is to determine how $T_{\mu\nu}$ fixes the curvature and hence the connection. The present lecture has assembled all the pieces that equation must respect: tensorial curvature, local conservation, and geodesic motion.


 A photograph of a chalkboard with the geodesic equation written in chalk. The equation is:

$$\frac{d}{d\tau} \frac{dx^\mu}{d\tau} + \Gamma^\mu_{\nu\sigma} \frac{dx^\nu}{d\tau} \frac{dx^\sigma}{d\tau} = 0$$
 The handwriting is in black chalk on a light-colored board. The equation is centered and written in a clear, legible style.

Figure 1.19: The completed geodesic equation. The connection turns the locally straight worldline into coordinate acceleration in a general chart.

Lecture frame: 01:52:23.

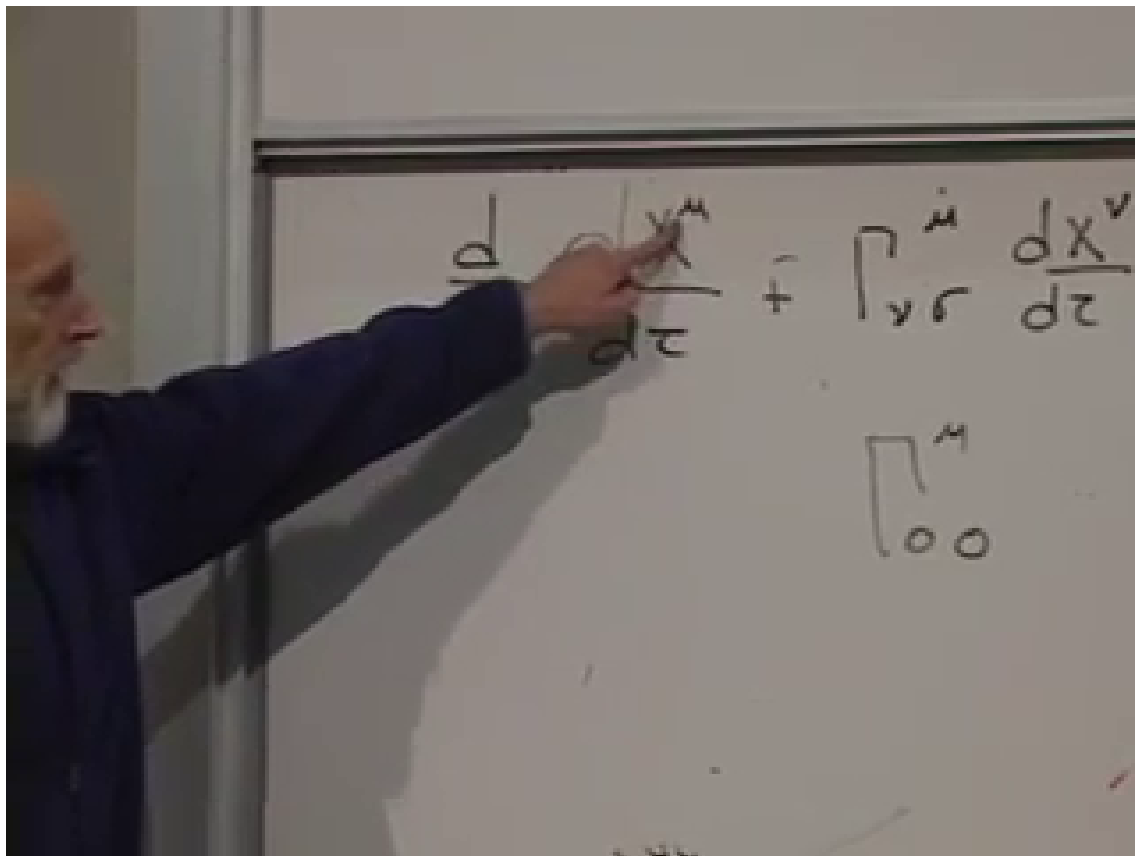


Figure 1.20: The geodesic equation beside the dominant weak-field coefficient Γ^μ_{00} . For spatial motion the acceleration is $-\Gamma^i_{00}$ in the stated convention.

Lecture frame: 01:55:38.